

ISAAC SHEIDLLOWER

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EDUCATION

Tufts University

PhD: Department of Computer Science

2020-2025

Advisor: Dr. Elaine Short

Rutgers University New Brunswick

Bachelor of the Arts

2016-2020

Major: Cognitive Science; Minor: Computer Science

Honors: Rutgers New Brunswick Honors Program, Phi Beta Kappa

PUBLICATIONS AND PAPERS

Conference Publications

Isaac Sheidlower, Emma Bethel, Douglas Lilly, Reuben M. Aronson and Elaine Schaertl Short, "Imagining In-distribution States: How Predictable Robot Behavior Can Enable User Control Over Learned Policies," 2024 33rd IEEE International Conference on Robot and Human Interactive Communication (RO-MAN), Pasadena, CA, USA, 2024, <https://ieeexplore.ieee.org/document/10731233>

Jindan Huang, Isaac Sheidlower, Reuben M Aronson, and Elaine Schaertl Short. 2024. On the Effect of Robot Errors on Human Teaching Dynamics. In Proceedings of the 12th International Conference on Human-Agent Interaction (HAI '24). Association for Computing Machinery, New York, NY, USA, 150–159. <https://doi.org/10.1145/3687272.3688320>

Isaac Sheidlower, Mavis Murdock, Emma Bethel, Reuben M. Aronson, and Elaine Short. "Online Behavior Modification for Expressive User Control of RL-Trained Robots." In *Proceedings of the 2024 ACM/IEEE International Conference on Human-Robot Interaction*, 639–48. HRI '24. New York, NY, USA: Association for Computing Machinery, 2024. <https://doi.org/10.1145/3610977.3634947>.

Isaac Sheidlower, Reuben Aronson, and Elaine Short (2024). Modifying RL Policies with Imagined Actions: How Predictable Policies Can Enable Users to Perform Novel Tasks. Proceedings of the AAAI Symposium Series, 2(1), 192-197. <https://doi.org/10.1609/aaais.v2i1.27670>

Isaac Sheidlower, Allison Moore and Elaine Short, "Keeping Humans in the Loop: Teaching via Feedback in Continuous Action Space Environments," 2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Kyoto, Japan, 2022, pp. 863-870, doi: 10.1109/IROS47612.2022.9982282. <https://ieeexplore.ieee.org/document/9982282>

Isaac Sheidlower, Allison Moore, and Elaine Short. "Environment Guided Interactive Reinforcement Learning: Learning from Binary Feedback in High-Dimensional Robot Task Environments." In Proceedings of the 21st International Conference on Autonomous Agents and Multiagent Systems, 1726–28. AAMAS '22, <https://dl.acm.org/doi/10.5555/3535850.3536090> (Extended Abstract)

Workshop Publications

Isaac Sheidlower, Reuben M. Aronson, and Elaine Short. "Towards Interpretable Foundation Models of Robot Behavior: A Task Specific Policy Generation Approach" (RLC 2024, Training Agents with Foundation Models) <https://openreview.net/pdf?id=umkqPTUDED>

Isaac S. Sheidlower and Elaine Schaertl Short. 2021. When Oracles Go Wrong: Using Preferences as a Means to Explore. In Companion of the 2021 ACM/IEEE International Conference on Human-Robot Interaction (HRI '21 Companion). Association for Computing Machinery, New York, NY, USA, 344–348. <https://doi.org/10.1145/3434074.3447189> **Best LBR Nominee.**

MENTORING

Tufts Undergraduate Students:

Tyler Chen & Bingyu Wu (September 2024-), working on projects related to user understanding of generalist robot policies
Douglas Lilly (September 2023-), worked on developing a flower bed with multiple weight sensors for detecting watering success
Emma Bethel (January 2023-June 2024), worked on developing novel assistive robot and anomaly detection algorithms
Alina Shah & Teo Patrosio (May 2022-June 2023), worked on developing a codebase for a Fetch Freight100
Allison Moore (Sep. 2021 – May 2022), worked on data visualization techniques for educational and research purposes

REVIEWER

Conferences: Int. Conference on Autonomous Agents and Multiagent Systems (AAMAS, Program Committee), International Joint Conferences on Artificial Intelligence (IJCAI, Program Committee), ACM/IEEE Conference on Human-Robot Interaction (HRI), ACM Transactions on Human-Robot Interaction (T-HRI)

TEACHING

Tufts Course Teaching:

Introduction to ROS – Spring 2023

Tufts Teaching Assistant:

Comp 150/ME 193: Assistive Algorithms - Spring 2021

RELATED EXPERIENCE

ACM/IEEE Conference on Human-Robot Interaction (HRI)

Student Volunteer**2021**

Assisted in editing camera-ready PDFs to be more accessible.